

1) list of variables and their datatypes in bus ticket. Task 1

Variable	Datatype
Ticket no	→ int
time	→ string
depot	→ string
date	→ string
route	→ string
place	→ string
balane	→ float
Total amount	→ float
vehicle no	→ string

2) current bill

service no	→ string
Consumer name	→ string
Address	→ string
phase	→ int
connected load	→ float
meter number	→ long
present reading	→ int
units	→ int
RMD	→ int
fixed charge	→ float

customer charge → float

GO → float

Bill amount → float

net bill → float

due date → string

bill date → string

time → string

section → string

district → string

QR code → image

3) Gas bill

Consumer number → long int

Cylinder price → double / float

sgst → double / float

cgst → float [double

customer name → string

final price → float / double

invoice date → string

order status → string

connection type → string

47 Amazon order

Billing Address $\rightarrow$ string

Shipping Address $\rightarrow$ string

pan no $\rightarrow$ string

gst no $\rightarrow$ string

order number $\rightarrow$ string

order date $\rightarrow$ string

invoice number $\rightarrow$ string

invoice date $\rightarrow$ string

product name $\rightarrow$ string

unit price $\rightarrow$ float

quantity $\rightarrow$ int

net amount $\rightarrow$ float

tax rate $\rightarrow$ float

tax amount $\rightarrow$ float

total amount $\rightarrow$ float

Payment transaction ID $\rightarrow$ string

payment mode $\rightarrow$ string

date $\rightarrow$ string

time $\rightarrow$ string

Invoice value $\rightarrow$ float

4) The Real-life Examples of DSA

• Queue - Students are waiting at Canteen
- counter

Stack - Books piled on your desk

Array - houses & shops in a row

Tree - hierarchy in my college

Graph - the road network itself! Roads
Connecting hostel, college, bus stop

Hash-table - my collage Id Card (or) roll
number system.

2D array / matrix - college timetable

9/7/20 programming Task-3

```
#include <stdio.h>

struct student
{
    int id;
    char grade;
    float marks;
};

int main()
{
    struct student s;
    printf("size of structure student = %d byte",
           sizeof(s));
    return 0;
}
```

manual size calculation.

Data member	Data type	size (bytes)
id	int	4
grade	char	1
marks	float	4
Total		9 bytes

Actual size using sizeof
size of structure student = 12 bytes

why are they different?

The outputs are different

- Manual size: 9 bytes

10/7/26

Task-4

① open amazon website find out 10 different functions

② Search return 0,1 means what the main returns to whom.

1) productId = int Add to cart
productName = string Buy now
productPrice = float Wishlist
quantity = int Add to your order
customerName = string Account
customerEmail = string Reviews
cartItems = array filter
shippingAddress = object Cancel order
paymentMethod = string
order status = string

2) Return 0 → indicates successful program execution and returns control to the OS.

Return 1 → indicates unsuccessful execution (or an error) and returns control to OS.

main() → returns to the OS.

• Size of result: 12 bytes (on most compiler).

This difference occurs because the compiler adds padding bytes b/w members to align data in memory for faster access.
memory layout.

member	size	offset
id	4 bytes	0-3
grade	1 "	4
padding	3 "	5-7
marks	4 "	8-11

So, the total size become 12 bytes instead of 9 bytes.

Hence, they are different because the compiler inserts padding to satisfy memory alignment requirements, improving performance.

The exact size may vary depending on the compiler and system architecture.

Algorithms:

- Searching: linear & binary
- Sorting - basic: Bubble, selection & insertion
- Sorting - advanced: merge, quick, heap sort
- Complexity: Asymptotic worst-case time and space complexity, Big-O, Omega, Theta
- Algorithm design techniques: Greedy, dynamic programming, divide-and-conquer
- Graph algorithms.

3) → Message to a friend

Inputs:

- Sender's name / number
- Receiver's name / number
- Message content
- Timestamp

Outputs:

- Delivered message
- Delivery status
- Timestamp of delivery

2) Bus ticket

Inputs

- present Station
- destination station
- Date of travel
- no of passengers
- Bus Type

Outputs

- Ticket number
- Seat number
- fare amount
- Departure time
- Destination point
- Total amount to paid

3) Electricity Bill

Inputs

- consumer ID
- previous meter reading
- current meter reading
- no of units consumed
- Rate per unit
- meter rent
- due date

Outputs

- units consumed
- Total Bill
- Due date for payment
- Late payment penalty

HW 14/7/26

Task-5

Compute the time complexity using the frequency count for the following sequence.

1) for ($i=1$; $i \leq 10$; $i=i+2$)
{
 $i=i \times 10$;
 printf ("%d", i);
}

= 22

2) $i=1$;
while ($i \leq 50$); ~~$do = 100$~~
{
 printf ("hello");
}

3) for ($i=1$; $i \leq 20$; $i++$);